

Smart Attendance System Using BLE Beacons & Flutter App

Milestone 1: Setup Basic Flutter App and BLE Beacons

Objective: Create a basic Flutter app to detect beacon signals.

Software Tools:

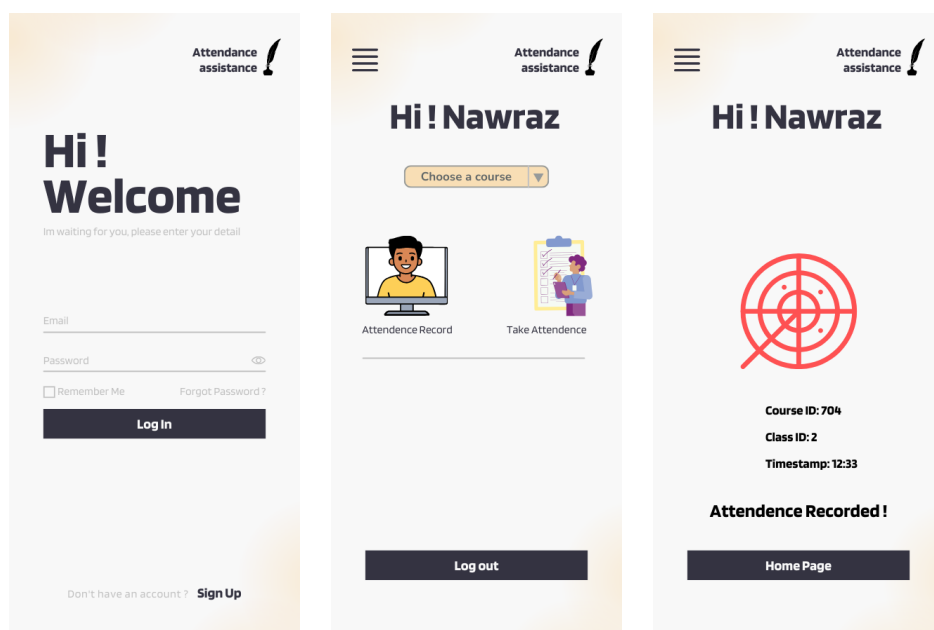
- You will be working with Android Studio IDE. You can find supporting materials on the cms.
- As you will create a Flutter app, you need to download and install Flutter SDK from the official [website](#).
- You can now create a new Flutter project on Android Studio.

Requirements:

1. **Build a Basic Flutter App :**

- (a) Implement user authentication using Firebase Authentication, allowing students to create accounts and log in securely.
- (b) Utilize email/password authentication or integrate with popular OAuth providers like Google or Facebook for streamlined sign-up and login processes.
- (c) Integrate a BLE scanning package (e.g., flutter_blue_plus).
- (d) Implement Bluetooth functionality to enable the Flutter app to scan for nearby BLE devices, including beacons.
- (e) Develop the logic to establish a connection with the beacon detected by the Flutter App.
- (f) Display the detected beacon's UUID and signal strength (RSSI) on the app screen.
- (g) calculate the approximate distance between the device and the beacon to decide if the status is attended.
- (h) Enable each student to monitor their attendance level at each class (Front-end only).

Main home page should be some thing like this :



You should Design and implement the user interface for the application, including screens for different options (Home page), beacon discovery (Take attendance), beacon connection status and distance detection (To define if attendance is recorded).

2. Deploy BLE Beacons :

- Assume that a beacon is already installed and powered in a tutorial room.
- You need to make sure that the beacon is configured to be assigned a unique UUID for identification.
- Make sure that the beacon's signal strength (RSSI) is adjusted to define the detection range (e.g., 5-10 meters)
- You can verify that the beacon is broadcasting their UUID correctly by a BLE Beacon scanner.

Team leaders should coordinate the testing of their team's Flutter app to ensure it detects beacons and displays their RSSI and UUID correctly

Submissions Guidelines :

- You can work in teams of 3 minimum and 5 maximum. Kindly submit your team info through this [link](#).
- The deadline for milestone 1 submission is 12-4-2025 at 11:59 pm.
- Any cheating cases will get a zero.
- you'll need to submit:
 - A working app.
 - Screenshots of the database.
- Only one team member (preferably team leader) should submit the required submissions through this [form](#).

Helping Materials :

- [Android Developer Documentation](#): (Official source for Android app development guides, tutorials, and APIs)
- [Stack Overflow](#): (Vast community forum for programming questions and answers)
- [Flutter Tutorials Center](#): (Documentation for the Flutter and dart programming language widely used in Android development)
- [flutter blue plus library](#): (Popular library for interacting with BLE beacons on Flutter apps)
- [Radius Networks Documentation](#): (Guides and tutorials for their beacon hardware and software platform)
- [Estimote Documentation](#): (Guides and tutorials for their beacon hardware and software platform)
- [BeaconZone](#): (Open-source framework for developing beacon-based apps)
- [Firebase Documentation](#):
- [Firebase Realtime Database or Cloud Firestore](#):
- [Android Authentication with Firebase](#):